1. ADRENAL INSUFFICIENCY & ADRENAL CRISIS

Primary (Addison) vs Secondary Insufficiency & Acute Crisis Resuscitation Protocol

PATHOPHYSIOLOGY & ETIOLOGY

Primary (Addison's): Autoimmune (80%) or infection (TB) causing destruction of adrenal cortex -> deficiency of cortisol, aldosterone, and androgens.
Secondary/Tertiary: Exogenous steroid withdrawal or pituitary disorder -> deficit of ACTH and cortisol, but *aldosterone remains intact* (regulated by RAAS).
Adrenal Crisis: Acute severe decompensation triggered by infection, trauma, surgery, or abrupt steroid cessation.

CLINICAL PRESENTATION

Chronic: Fatigue, weakness, anorexia, weight loss, dizziness, hyperpigmentation (palmar creases, buccal mucosa in primary due to ACTH/MSH surge).
Acute Crisis: Severe refractory hypotension/shock, high fever, severe abdominal/back pain, confusion, and vomiting.

PHYSICAL EXAMINATION

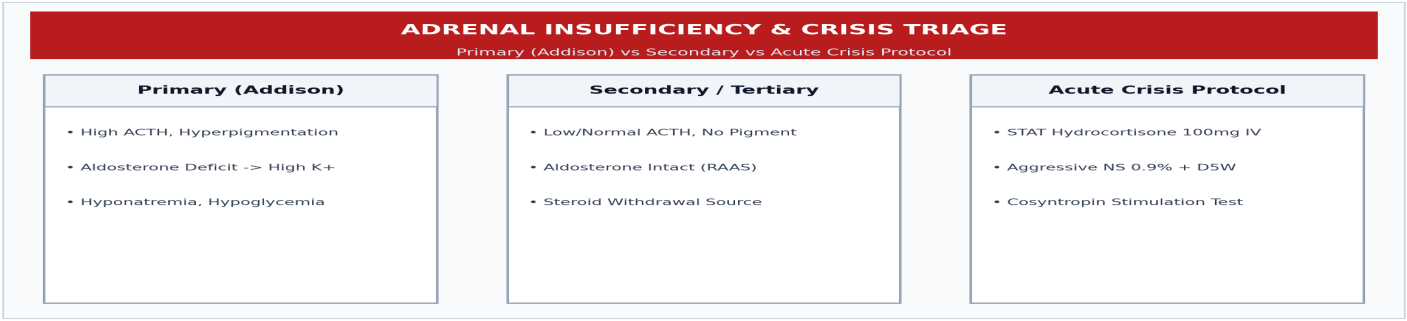
Hypertension vs Shock, generalized hyperpigmentation (Primary only), orthostatic hypotension, confusion, flank/abdominal tenderness.

DIAGNOSTIC STUDIES

Labs: Hyponatremia, Hyperkalemia (Primary only), Hypoglycemia, Hypercalcemia, Azotemia.
Diagnostic: Low 8 AM Serum Cortisol (<3 ug/dL). Confirmed by Cosyntropin (ACTH) Stimulation Test (failure of cortisol to rise >18 ug/dL).

MANAGEMENT & TREATMENT

Acute Adrenal Crisis:
1. **STAT Hydrocortisone:** 100 mg IV bolus immediately (does not interfere with ACTH stim test).
2. **Fluid Resuscitation:** 1-2L Normal Saline 0.9% + Dextrose 5% (D5NS) for hypoglycemia & hypotension.
3. Treat underlying precipitants & mineralocorticoid (Fludrocortisone 0.1 mg) once oral intake restored.



HIGH-YIELD BOARD PEARL:

In acute adrenal crisis, NEVER delay steroid administration for diagnostic testing! Hydrocortisone 100mg IV provides both glucocorticoid and mineralocorticoid activity. Dexamethasone 4mg IV can be used if immediate ACTH stimulation testing is planned, as it does not cross-react with cortisol assays.

2. CUSHING DISEASE & CUSHING SYNDROME

Hypercortisolemia Etiology, Clinical Features & Diagnostic Localization Cascade

PATHOPHYSIOLOGY & ETIOLOGY

Cushing Syndrome: State of chronic hypercortisolemia. Most common overall cause is *Exogenous iatrogenic corticosteroid therapy*.
Cushing Disease: Endogenous hypercortisolemia caused specifically by an ACTH-secreting pituitary adenoma (70% of endogenous cases).
Ectopic ACTH: ACTH production by non-pituitary tumors (Small Cell Lung Cancer).

CLINICAL PRESENTATION

Progressive central obesity, moon facies, buffalo hump, purple abdominal striae, easy bruising, muscle weakness, hirsutism, hypertension, glucose intolerance, amenorrhea, and psychiatric lability.

PHYSICAL EXAMINATION

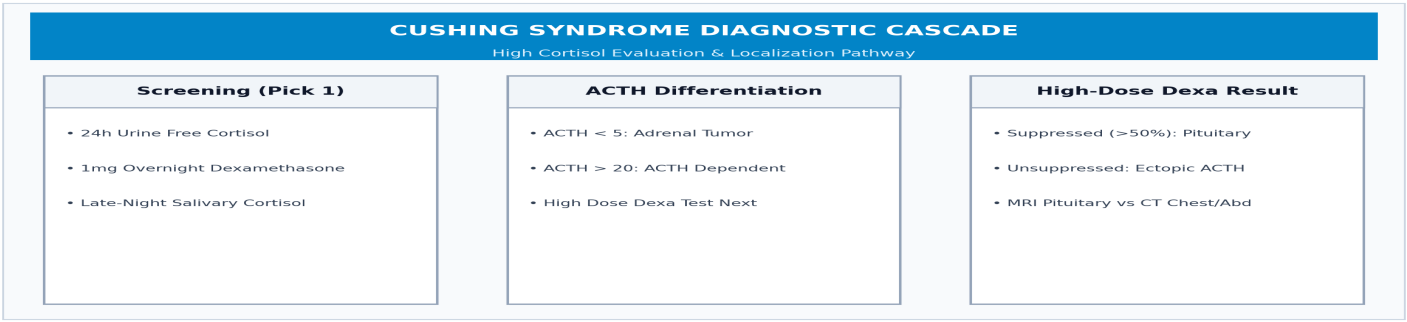
Plethoric moon-shaped face, supraclavicular fat pads, thin skin with wide (>1cm) purple striae, proximal muscle weakness (difficulty standing from chair).

DIAGNOSTIC STUDIES

Screening Tests (Pick 1):
1. 24-hour Urine Free Cortisol (elevated).
2. 1mg Overnight Dexamethasone Suppression Test (failure to suppress cortisol <1.8 ug/dL).
3. Late-night Salivary Cortisol.
Localization: Plasma ACTH. If high -> High-dose (8mg) Dexa Suppression Test (suppression >50% = Pituitary Cushing Disease; no suppression = Ectopic SCLC).

MANAGEMENT & TREATMENT

Exogenous: Taper corticosteroids slowly.
Pituitary Adenoma: Transsphenoidal Surgical Resection (first-line).
Adrenal Tumor: Laparoscopic Adrenalectomy.
Ectopic ACTH: Resection of underlying malignancy; Medical suppression (Ketoconazole, Metyrapone) if unresectable.



HIGH-YIELD BOARD PEARL:

Exogenous steroid administration is the most common cause of Cushing syndrome. When evaluating endogenous Cushing, ACTH levels differentiate adrenal causes (low ACTH) from pituitary/ectopic causes (high ACTH). High-dose dexamethasone suppresses pituitary ACTH but NOT ectopic ACTH.

3. DIABETES INSIPIDUS (CENTRAL VS NEPHROGENIC)

Polyuria Etiology, Water Deprivation Protocol & Desmopressin Triage

PATHOPHYSIOLOGY & ETIOLOGY

Inability to concentrate urine resulting in severe polyuria (>3L/24h) and polydipsia.
Central DI: Deficient synthesis/release of ADH (Vasopressin) from posterior pituitary due to head trauma, neurosurgery, or tumors.
Nephrogenic DI: Renal resistance to circulating ADH due to Lithium toxicity, hypercalcemia, hypokalemia, or congenital AVPR2 mutation.

CLINICAL PRESENTATION

Intense unquenchable thirst (predilection for ice-cold water), massive high-volume dilute urine output (up to 20L/day), nocturia, dehydration, and hypernatremic confusion.

PHYSICAL EXAMINATION

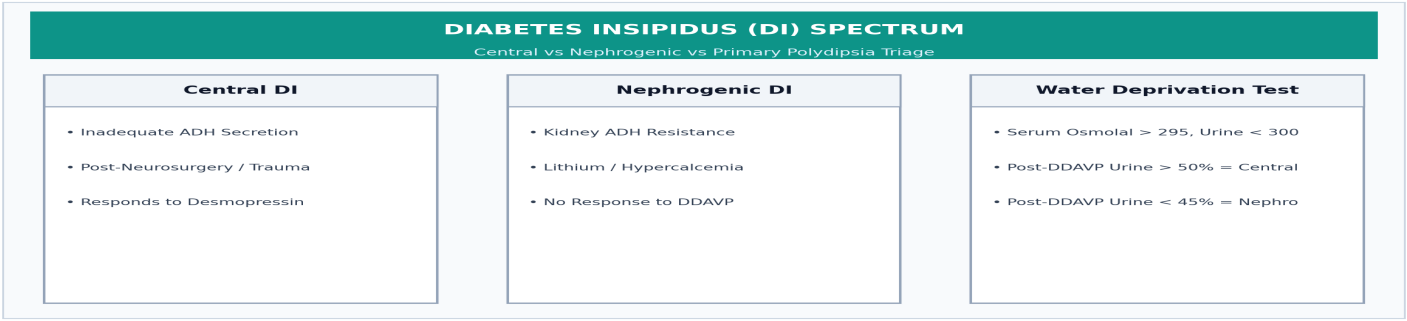
Signs of severe volume depletion (tachycardia, dry mucous membranes, decreased skin turgor, orthostasis).

DIAGNOSTIC STUDIES

Initial Serum/Urine: High Serum Osmolality (>295 mOsm/kg), High Serum Sodium (>145 mEq/L), Low Urine Osmolality (<300 mOsm/kg), Low Urine Specific Gravity (<1.005).
Water Deprivation Test: Urine remains dilute despite dehydration.
Desmopressin (DDAVP) Challenge:
• *Central DI:* Urine Osmolality increases by >50%.
• *Nephrogenic DI:* Minimal or no increase in Urine Osmolality (<45%).

MANAGEMENT & TREATMENT

Central DI: Desmopressin (DDAVP) intranasal or oral (drug of choice). Free water replacement.
Nephrogenic DI: Stop offending agent (Lithium). Thiazide diuretics + Indomethacin + Low-sodium diet.
Acute Hypernatremia: Calculate free water deficit; replace slowly with D5W to prevent cerebral edema.



HIGH-YIELD BOARD PEARL:

A patient with dilute urine (Urine Osm < 300) in the setting of elevated serum osmolality (>295) has Diabetes Insipidus. Desmopressin (DDAVP) administration corrects Central DI (Urine Osm rises >50%) but has no effect on Nephrogenic DI.

4. DIABETES MELLITUS MANAGEMENT & COMPLICATIONS

Type 1 vs Type 2 Emergency Diagnostic Parameters & Micro/Macrovascular Target Care

PATHOPHYSIOLOGY & ETIOLOGY

Type 1 DM: Autoimmune destruction of pancreatic beta cells -> absolute insulin deficiency.
Type 2 DM: Progressive insulin resistance combined with secretory defect.
Complications: Non-enzymatic glycation of tissue proteins leading to microvascular damage (retinopathy, nephropathy, neuropathy) and accelerated macrovascular atherosclerosis.

CLINICAL PRESENTATION

Polyuria, polydipsia, polyphagia, unintentional weight loss, blurred vision, slow-healing wounds, peripheral numbness/tingling, and recurrent infections.

PHYSICAL EXAMINATION

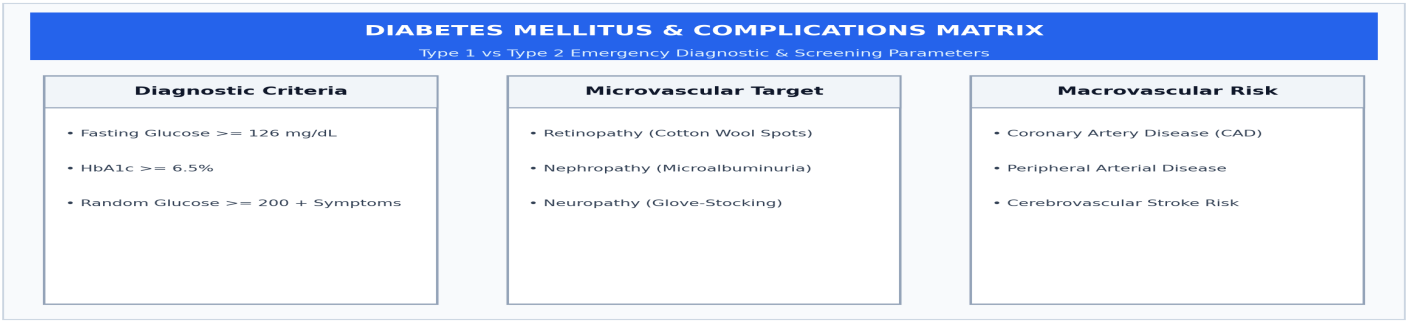
Glove-and-stocking sensory loss, absent ankle reflexes, neuropathic ulcers, diabetic retinopathy (cotton wool spots, neovascularization), acanthosis nigricans.

DIAGNOSTIC STUDIES

Diagnostic Criteria (Any of 4):
1. Fasting Plasma Glucose >= 126 mg/dL.
2. 2-hour Post-load Glucose >= 200 mg/dL during OGTT.
3. HbA1c >= 6.5%.
4. Random Glucose >= 200 mg/dL with classic hyperglycemic symptoms.
Microalbuminuria: Urine albumin-to-creatinine ratio >= 30 mg/g.

MANAGEMENT & TREATMENT

Type 1 DM: Basal-bolus insulin regimen.
Type 2 DM: Lifestyle modifications + Metformin (1st line). Add SGLT2 inhibitors or GLP-1 receptor agonists if CKD, HF, or ASCVD present.
Hypertension/CKD: ACE Inhibitors or ARBs (first line for renal protection).
Lipids: Moderate-to-high intensity Statin for all diabetics aged 40-75.



HIGH-YIELD BOARD PEARL:

SGLT2 inhibitors (e.g., Empagliflozin) reduce mortality, heart failure hospitalizations, and CKD progression in diabetics. Note: SGLT2 inhibitors can cause Euglycemic DKA! Metformin must be held before IV contrast administration in patients with eGFR < 30-45.

5. DIABETIC KETOACIDOSIS (DKA) PROTOCOL

Hyperglycemic Crisis Resuscitation, Insulin Infusion & Anion Gap Closure Target

PATHOPHYSIOLOGY & ETIOLOGY

Absolute insulin deficiency combined with glucagon excess -> severe hyperglycemia, osmotic diuresis, lipolysis, and hepatic ketogenesis (beta-hydroxybutyrate and acetoacetate) resulting in high anion gap metabolic acidosis.

CLINICAL PRESENTATION

Nausea, vomiting, abdominal pain (may mimic acute abdomen), polyuria, polydipsia, lethargy, confusion, Kussmaul respiration (deep, rapid breathing), and fruity/acetone breath odor.

PHYSICAL EXAMINATION

Tachycardia, hypotension, dry mucous membranes, fruity odor, altered sensorium, diffuse abdominal tenderness without peritoneal signs.

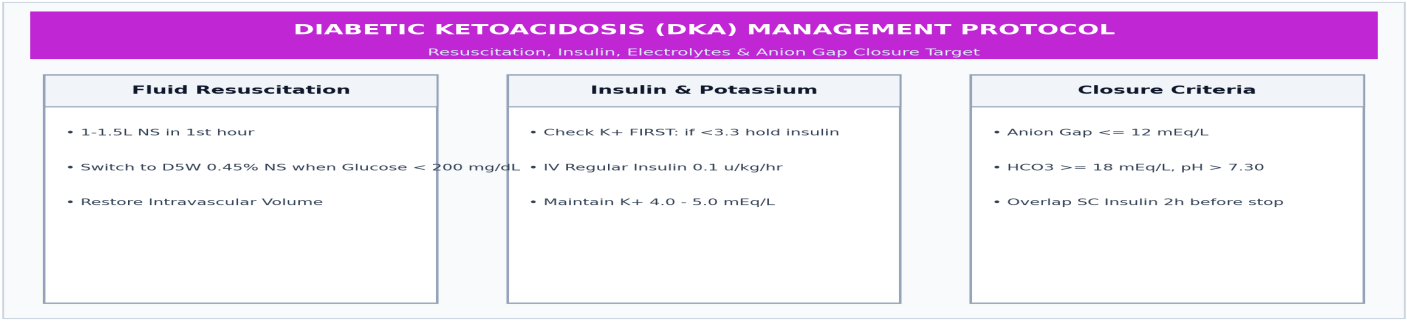
DIAGNOSTIC STUDIES

Diagnostic Triad:

1. Serum Glucose > 250 mg/dL.
2. Arterial pH < 7.30 or Serum Bicarbonate < 18 mEq/L.
3. High Serum Ketones (Beta-hydroxybutyrate) & High Anion Gap (>12).

MANAGEMENT & TREATMENT

1. Fluids First: 1-1.5L Normal Saline (0.9% NaCl) in 1st hour.
2. Potassium Check:
• If K+ < 3.3 mEq/L: **HOLD INSULIN**, give IV Potassium until K+ > 3.3.
• If K+ 3.3-5.3: Add 20-30 mEq K+ per liter of IV fluid.
3. Insulin: Regular Insulin IV 0.1 units/kg/hr infusion.
4. Dextrose Addition: Add D5W to IV fluids when glucose reaches 200 mg/dL.
5. Resolution Criteria: Glucose < 200, HCO3 >= 18, pH > 7.30, Anion Gap <= 12.



HIGH-YIELD BOARD PEARL:

ALWAYS check Potassium before starting insulin in DKA! Insulin shifts potassium intracellularly and can cause fatal dysrhythmias if K+ < 3.3 mEq/L. Switch to D5W when glucose drops < 200 mg/dL while continuing insulin infusion to clear ketoacidosis without causing hypoglycemia.

6. HEAT & COLD INTOLERANCE (ENDOCRINE TRIAGE)

Thermoregulatory Disruption Differential: Thyrotoxicosis vs Myxedema Coma

PATHOPHYSIOLOGY & ETIOLOGY

Thyroid hormones directly regulate basal metabolic rate (BMR) and Na+/K+ ATPase activity.
Heat Intolerance: Hyperthyroidism causes uncoupled oxidative phosphorylation, excess thermogenesis, and beta-adrenergic overactivity.
Cold Intolerance: Hypothyroidism causes low BMR, decreased heat production, and metabolic slowing.

CLINICAL PRESENTATION

Heat Intolerance: Excessive sweating, palpitations, weight loss despite increased appetite, tremor, diarrhea, insomnia, and anxiety.
Cold Intolerance: Constant cold sensation, lethargy, weight gain, constipation, dry skin, cold extremities, and hoarseness.

PHYSICAL EXAMINATION

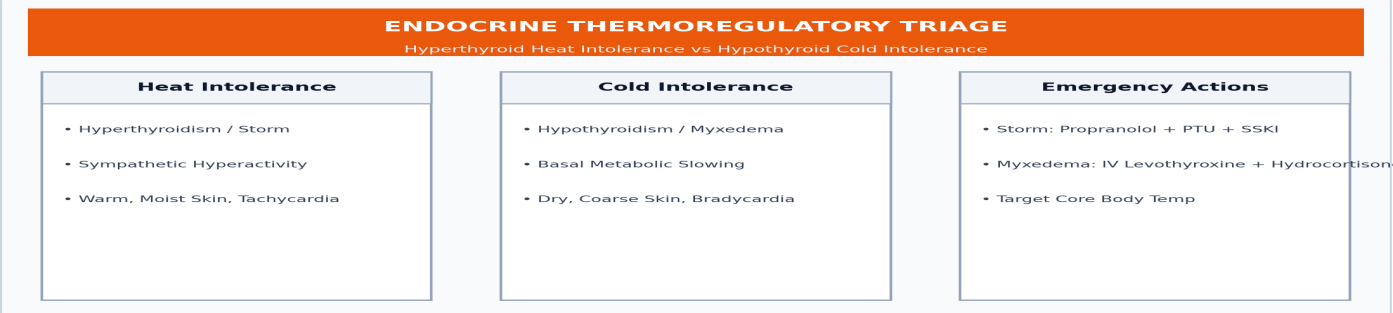
Hyperthyroid: Tachycardia, moist/warm skin, fine tremor, hyperreflexia.
Hypothyroid: Bradycardia, hypothermia, dry/coarse skin, delayed relaxation phase of deep tendon reflexes (hung-up reflexes).

DIAGNOSTIC STUDIES

Serum TSH and Free T4 levels.
• High T4 + Low TSH = Primary Hyperthyroidism.
• Low T4 + High TSH = Primary Hypothyroidism.
ECG: Atrial Fibrillation in hyperthyroid; Bradycardia/Low voltage in hypothyroid.

MANAGEMENT & TREATMENT

Heat Intolerance / Thyrotoxicosis: Propranolol (symptomatic relief) + Methimazole / PTU.
Cold Intolerance / Hypothyroidism: Oral Levothyroxine (T4) replacement.
Severe Extremes: Active cooling for Thyroid Storm; Active rewarming + IV Levothyroxine for Myxedema Coma.



HIGH-YIELD BOARD PEARL:

Unexplained heat intolerance with tachycardia demands immediate evaluation for thyrotoxicosis. Unexplained cold intolerance with hypothermia and altered mental status in an elderly patient during winter suggests Myxedema Coma.

7. HYPERPARATHYROIDISM & HYPERCALCEMIA CRISIS

Parathyroid Adenoma, 'Bones, Stones, Groans, Psychiatric Overtones' & Emergency Resuscitation

PATHOPHYSIOLOGY & ETIOLOGY

Primary Hyperparathyroidism: Autonomous PTH oversecretion (85% solitary parathyroid adenoma) -> increased bone resorption, renal Ca2+ reabsorption, and vitamin D synthesis.
Hypercalcemia Crisis: Serum Ca2+ > 14 mg/dL causing severe dehydration, acute renal failure, and altered mental status.

CLINICAL PRESENTATION

Mnemonic: **'Bones, Stones, Groans, Psychiatric Overtones':**
• *Bones:* Bone pain, fractures (Osteitis fibrosa cystica).
• *Stones:* Nephrolithiasis, nephrocalcinosis.
• *Groans:* Abdominal pain, constipation, nausea, pancreatitis.
• *Psychiatric:* Lethargy, depression, confusion, coma.

PHYSICAL EXAMINATION

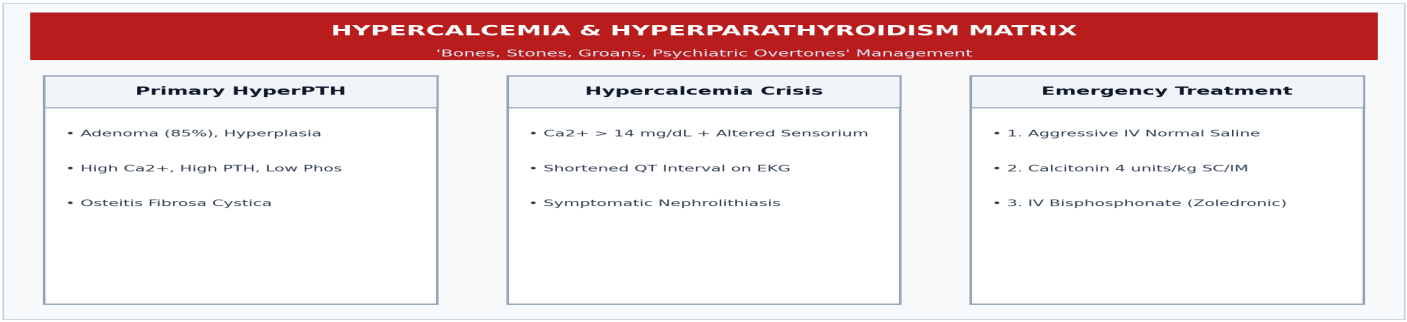
Muscle weakness, hyporeflexia, dehydration, confusion, abdominal distension.

DIAGNOSTIC STUDIES

Serum Labs: High Ca2+, High PTH, Low Phosphate, High 24h Urine Ca2+.
ECG: Shortened QT Interval, PR prolongation, Bradycardia.
Imaging: Sestamibi Scan for adenoma localization; Bone density (DEXA).

MANAGEMENT & TREATMENT

Hypercalcemia Crisis (Ca2+ > 14 or Symptomatic):
1. **Volume Resuscitation:** Aggressive IV Normal Saline (0.9% NaCl) 200-500 mL/hr.
2. **Calcitonin:** 4 units/kg SC/IM (rapid onset 2-4h, temporary).
3. **Bisphosphonates:** IV Zoledronic Acid (4mg) or Pamidronate (onset 24-48h).
4. **Dialysis:** Refractory cases or severe renal/heart failure.
Definitive: Parathyroidectomy.



HIGH-YIELD BOARD PEARL:

Aggressive IV Normal Saline hydration is the single most important initial therapy for hypercalcemic crisis! Avoid loop diuretics (Furosemide) UNTIL volume resuscitation is complete. Calcitonin works fast (hours); Bisphosphonates provide durable control (days).

8. HYPERTHYROIDISM & GRAVE'S DISEASE

Thyrotoxicosis Etiology, Eye & Skin Manifestations & Antithyroid Drug Triage

PATHOPHYSIOLOGY & ETIOLOGY

Grave's Disease: Autoimmune disease caused by Thyroid-Stimulating Immunoglobulin (TSI) activating TSH receptors -> diffuse hyperplasia.
Other Causes: Toxic Multinodular Goiter, Toxic Adenoma, Subacute Thyroiditis, Exogenous Levothyroxine abuse.

CLINICAL PRESENTATION

Palpitations, weight loss, heat intolerance, anxiety, tremors, frequent bowel movements, oligomenorrhea, exophthalmos (Grave's specific), pretibial myxedema.

PHYSICAL EXAMINATION

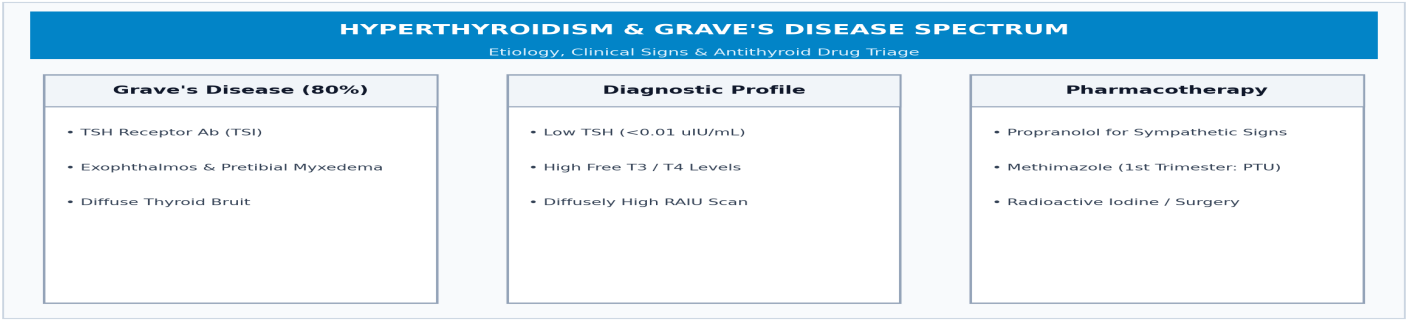
Tachycardia/AFib, fine finger tremor, warm/moist skin, diffuse smooth goiter with bruit (Grave's), lid lag, proptosis, brisk deep tendon reflexes.

DIAGNOSTIC STUDIES

Labs: Suppressed TSH (<0.01 uIU/mL), High Free T4 and Free T3.
Serology: Positive TSI / TSH-receptor antibodies.
Radioactive Iodine Uptake (RAIU):
• *Grave's:* Diffuse high uptake.
• *Toxic Nodular:* Nodular high uptake.
• *Thyroiditis:* Low/zero uptake.

MANAGEMENT & TREATMENT

Symptom Control: Propranolol (controls tachycardia and tremor).
Antithyroid Drugs (ATDs):
• **Methimazole:** First-line for most patients (lower hepatotoxicity).
• **Propylthiouracil (PTU):** Preferred in 1st trimester of pregnancy & Thyroid Storm.
Definitive: Radioactive Iodine (RAI) ablation or Total Thyroidectomy.



HIGH-YIELD BOARD PEARL:

Methimazole is the antithyroid drug of choice EXCEPT in the first trimester of pregnancy (where PTU is used due to Methimazole embryopathy risk) and in Thyroid Storm (where PTU is preferred because it also inhibits peripheral T4 to T3 conversion).

9. HYPOTHYROIDISM & MYXEDEMA COMA

Hashimoto Thyroiditis to Life-Threatening Myxedema Emergency Protocol

PATHOPHYSIOLOGY & ETIOLOGY

Primary Hypothyroidism: Hashimoto Thyroiditis (autoimmune anti-TPO/anti-thyroglobulin destruction) or Iatrogenic (post-RAI/surgery).
Myxedema Coma: Severe decompensated hypothyroidism precipitated by infection, cold exposure, trauma, or CNS depressants in long-standing untreated patients.

CLINICAL PRESENTATION

Hypothyroidism: Fatigue, cold intolerance, weight gain, constipation, dry skin, hair thinning, memory loss, menorrhagia, hoarseness.
Myxedema Coma: Altered mental status, obtundation, severe hypothermia (<35C/95F), hypoventilation, bradycardia, and generalized non-pitting edema.

PHYSICAL EXAMINATION

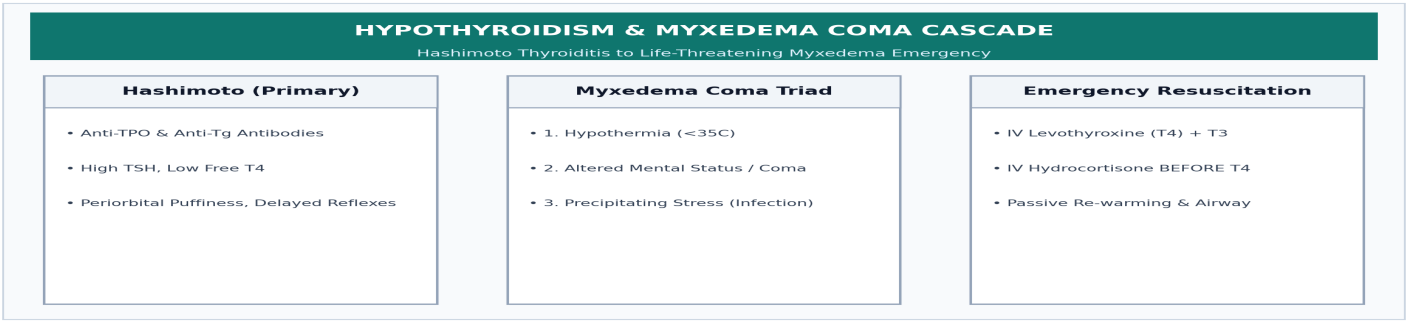
Periorbital puffiness, coarse dry skin, macroglossia, delayed relaxation of reflexes (hung-up reflexes), bradycardia, hypothermia, pericardial effusion.

DIAGNOSTIC STUDIES

Labs: High TSH, Low Free T4.
Myxedema: Severe hyponatremia, hypoglycemia, hypercapnic respiratory acidosis.
ECG: Sinus bradycardia, low voltage QRS, QT prolongation, flattened T waves.

MANAGEMENT & TREATMENT

Routine Hypothyroidism: Levothyroxine (T4) 1.6 mcg/kg/day PO.
Myxedema Coma Emergency:
1. **Airway & Passive Warming:** Mechanical ventilation; blankets (avoid rapid active rewarming -> collapse).
2. **IV Glucocorticoids FIRST:** Hydrocortisone 100mg IV (prevents adrenal crisis).
3. **IV Levothyroxine:** Loading dose 200-400 mcg IV, then 50-100 mcg daily.
4. Treat underlying infection & cautious isotonic fluids.



HIGH-YIELD BOARD PEARL:

In Myxedema Coma, ALWAYS administer IV Hydrocortisone BEFORE or concurrently with IV Levothyroxine! Giving thyroid hormone alone accelerates metabolism and can precipitate fatal adrenal crisis if co-existing adrenal insufficiency is present.

10. HYPEROSMOLAR HYPERGLYCEMIC STATE (HHS)

Severe Hyperglycemia, Hyperosmolality & Volume Resuscitation Protocol

PATHOPHYSIOLOGY & ETIOLOGY

Relative insulin deficiency and resistance in Type 2 DM -> extreme hyperglycemia and osmotic diuresis without significant ketogenesis (enough insulin present to suppress lipolysis/ketones, but not enough to prevent hyperglycemia).

CLINICAL PRESENTATION

Insidious onset over days to weeks: severe polyuria, polydipsia, profound weakness, lethargy, focal neurological deficits (hemiparesis, seizures), and coma.

PHYSICAL EXAMINATION

Severe dehydration (dry membranes, sunken eyes), orthostatic shock, altered sensorium, focal neuro deficits (may mimic stroke).

DIAGNOSTIC STUDIES

Diagnostic Profile:

1. Serum Glucose > 600 mg/dL (often >1000 mg/dL).
2. Effective Serum Osmolality > 320 mOsm/kg.
3. Arterial pH > 7.30 & Serum Bicarbonate > 18 mEq/L (No severe ketoacidosis).
4. Minimal/Absent Urine & Serum Ketones.

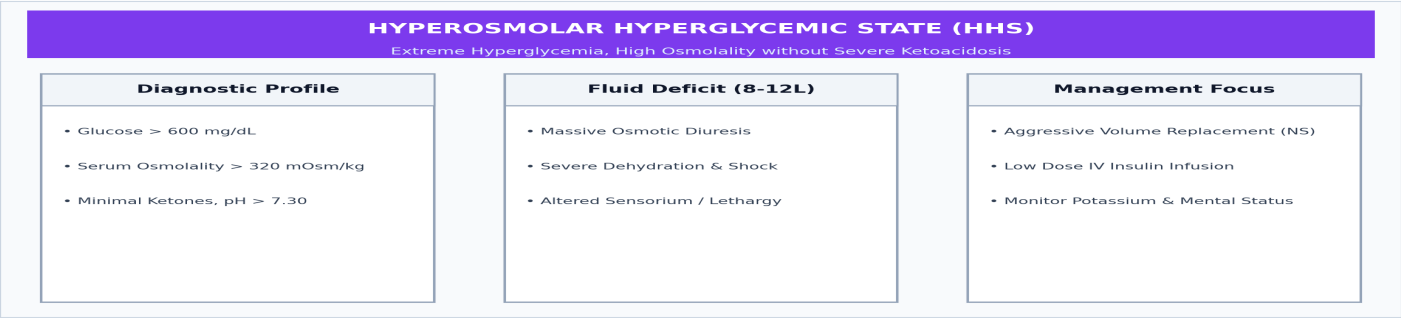
MANAGEMENT & TREATMENT

1. Massive Fluid Resuscitation (Primary Focus):

- Total fluid deficit is 8-12 Liters!
- 1-1.5L Normal Saline (0.9%) in 1st hour.
- Switch to 0.45% NaCl once hemodynamically stable.

2. Potassium Management: Maintain K+ 4.0-5.0 mEq/L.

3. Low-Dose IV Insulin: Regular Insulin 0.05-0.1 u/kg/hr (only after fluid resuscitation is underway). Add D5W when glucose reaches 250-300 mg/dL.



HIGH-YIELD BOARD PEARL:

HHS has a much higher mortality rate than DKA due to extreme dehydration (8-12L deficit) and underlying comorbidities in older patients. Fluid resuscitation is the single most critical intervention; insulin is secondary!

11. ENDOCRINE PALPITATIONS & THYROTOXICOSIS

Thyrotoxicosis, Pheochromocytoma & Hypoglycemic Surge Triage

PATHOPHYSIOLOGY & ETIOLOGY

Palpitations resulting from endocrine hyperadrenergic states:
1. **Thyrotoxicosis:** Increased beta-receptor expression & sensitivity.
2. **Pheochromocytoma:** Catecholamine-secreting tumor of chromaffin cells.
3. **Hypoglycemia:** Counter-regulatory epinephrine/sympathetic surge.

CLINICAL PRESENTATION

Thyrotoxic: Tachycardia, AFib, anxiety, heat intolerance.
Pheochromocytoma Classic Triad: Episodic Headache, Sweating, and Tachycardia with severe HTN.
Hypoglycemic: Tremor, diaphoresis, hunger, palpitations, confusion.

PHYSICAL EXAMINATION

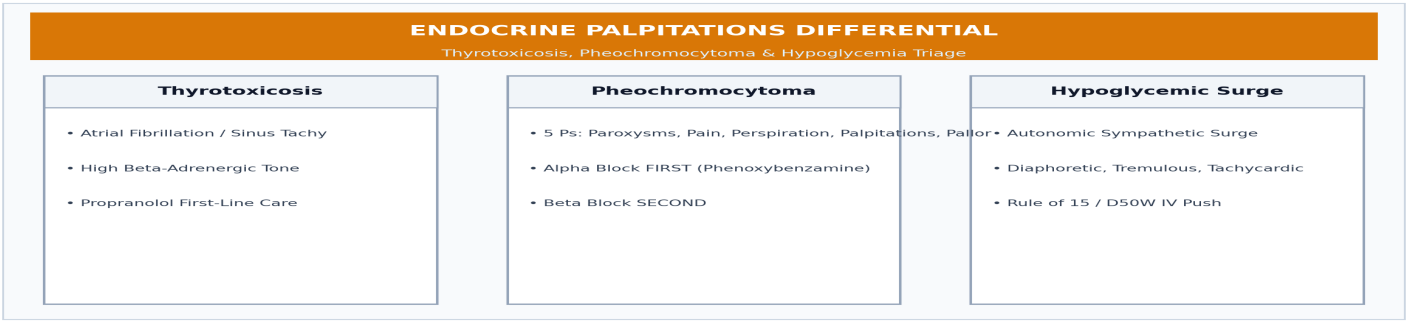
Pheo: Severe hypertension, paroxysmal pallor/flushing.
Hypoglycemia: Diaphoretic, tachycardic, altered sensorium.
Thyroid: Irregularly irregular pulse (AFib), goiter, tremor.

DIAGNOSTIC STUDIES

Thyroid: TSH, Free T4.
Pheochromocytoma: Plasma Free Metanephrines or 24h Urine Fractionated Metanephrines & Catecholamines. CT/MRI Abdomen.
Hypoglycemia: Bedside Fingerstick Glucose (<70 mg/dL).

MANAGEMENT & TREATMENT

Thyrotoxicosis: Propranolol (blocks beta-1 receptors & T4->T3 conversion).
Pheochromocytoma Pre-op Protocol:
1. **Alpha-Blocker FIRST:** Phenoxybenzamine or Doxazosin for 10-14 days.
2. **Beta-Blocker SECOND:** Propranolol added ONLY after adequate alpha-blockade.
Hypoglycemia: Rule of 15 (15g oral glucose) or D50W 1 amp (50 mL) IV push.



HIGH-YIELD BOARD PEARL:

In Pheochromocytoma, NEVER give a beta-blocker before an alpha-blocker! Unopposed alpha-stimulation will lead to hypertensive crisis, myocardial infarction, and flash pulmonary edema. Always 'Alpha before Beta'!

12. THYROIDITIS & THYROID STORM PROTOCOL

Life-Threatening Thyrotoxic Crisis Emergency 4-Step Multimodal Blockade

PATHOPHYSIOLOGY & ETIOLOGY

Thyroiditis: Inflammation of thyroid gland (Subacute De Quervain = painful viral; Hashimoto = painless autoimmune).
Thyroid Storm: Extreme, life-threatening hyperthyroidism decompensation triggered by infection, surgery, trauma, or RAI in hyperthyroid patients.

CLINICAL PRESENTATION

De Quervain Thyroiditis: Tender thyroid gland, fever, elevated ESR following viral upper respiratory infection.
Thyroid Storm: Severe high fever (104-106F), extreme tachycardia/AFib, agitation, delirium, psychosis, jaundice, and high-output heart failure.

PHYSICAL EXAMINATION

Burch-Wartofsky Point Scale >= 45. Marked fever, delirium, jaundice, pulmonary edema, severe tachycardia/AFib.

DIAGNOSTIC STUDIES

Clinical Diagnosis! Do not wait for lab confirmation.
Suppressed TSH, massively elevated Free T3/T4, leukocytosis, elevated LFTs.

MANAGEMENT & TREATMENT

Thyroid Storm Emergency 4-Step Blockade (GIVE IN EXACT ORDER):
1. **Step 1 (Beta Blockade):** Propranolol 60-80mg PO q6h or IV (controls adrenergic surge & T4->T3 conversion).
2. **Step 2 (Antithyroid Drug):** PTU 500-1000mg loading dose, then 250mg q4h (inhibits de novo synthesis).
3. **Step 3 (Iodine Blockade):** SSKI 5 drops PO or Lugol solution **GIVEN 1 HOUR AFTER PTU** (Wolff-Chaikoff effect).
4. **Step 4 (Glucocorticoids):** Hydrocortisone 100mg IV q8h (treats relative adrenal crisis & blocks T4->T3).

THYROID STORM EMERGENCY 4-STEP PROTOCOL
Life-Threatening Thyrotoxic Crisis Multimodal Blockade

Step 1: Beta-Blockade	Step 2: Antithyroid Drug	Step 3: Organified Iodine	Step 4: Steroid Support
• Propranolol IV / PO • Blocks Sympathetic Surge & Peripheral T4->T3 Conversion	• PTU 600-1000mg PO / Methimazole • Inhibits De Novo Hormone Synthesis	• SSKI / Lugol 1 Hour AFTER PTU • Wolff-Chaikoff Effect (Blocks Release)	• Hydrocortisone 100mg IV • Treats Relative Adrenal Insufficiency

HIGH-YIELD BOARD PEARL:

In Thyroid Storm, Organified Iodine (SSKI/Lugol) MUST be administered AT LEAST 1 HOUR AFTER PTU/Methimazole! If iodine is given first, it will serve as a substrate for new thyroid hormone synthesis and worsen the storm.

13. ENDOCRINE TREMORS & METABOLIC ENCEPHALOPATHY

Action/Postural Tremor Etiology, Workup & Metabolic Triage

PATHOPHYSIOLOGY & ETIOLOGY

Enhanced physiologic tremor caused by hyperadrenergic states, metabolic derangements, or toxic exposures causing increased motor unit synchronization.
Etiologies: Thyrotoxicosis, Hypoglycemia, Pheochromocytoma, Alcohol/Sedative Withdrawal, Caffeine, Beta-agonists, Steroids.

CLINICAL PRESENTATION

Postural or action tremor (worse with extended arms or goal-directed movement), accompanied by nervousness, diaphoresis, palpitations, or altered sensorium.

PHYSICAL EXAMINATION

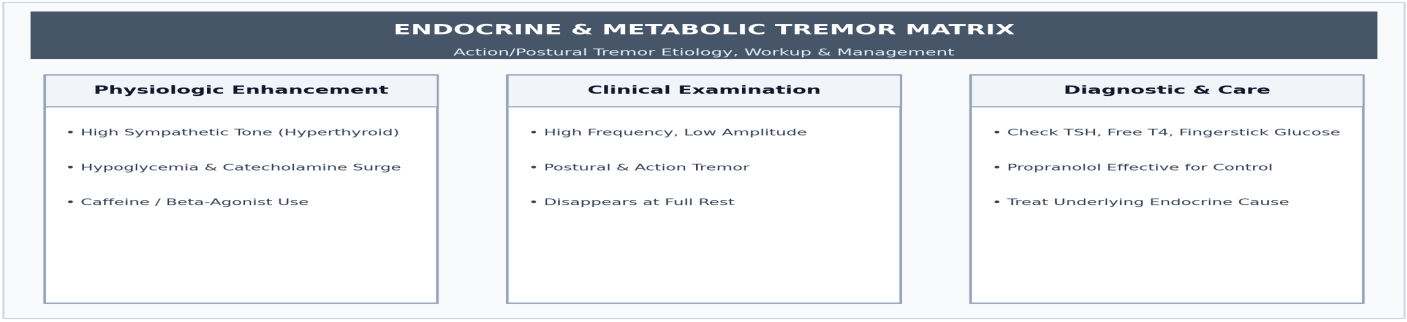
Fine, high-frequency, low-amplitude postural tremor of bilateral hands. Disappears when hands are fully supported at rest.

DIAGNOSTIC STUDIES

Fingerstick Glucose, TSH, Free T4, Serum Electrolytes, BUN/Creatinine, Liver Function Tests, Toxicology Screen.
EMG/Tremorogram if organic movement disorder suspected.

MANAGEMENT & TREATMENT

1. Treat Underlying Endocrine Cause: Correct hypoglycemia, manage thyrotoxicosis.
2. Symptomatic Relief: Propranolol 20-80mg PO daily (drug of choice for enhanced physiologic tremor).
3. Remove Triggers: Eliminate caffeine, sympathomimetics, and adjust tremor-inducing medications.



HIGH-YIELD BOARD PEARL:

An acute fine postural tremor accompanied by tachycardia and diaphoresis warrants immediate fingerstick glucose checking to exclude hypoglycemia, followed by TSH testing for hyperthyroidism. Propranolol provides prompt symptomatic relief.